Science practice booklet: Grade 8 expectations

Grade 8 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 8 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: MS-LS2-1, MS-LS2-2, MS-LS2-3, MS-LS2-4, MS-LS4-1, MS-LS4-2, MS-LS4-4

A worked example

In a mangrove ecosystem, algae are eaten by small fish and the small fish are eaten by herons. The algae capture 5000 units of energy from sunlight. About 10 percent of the energy is passed on at each step up the chain. How much energy reaches the herons, and where does the rest go?

1. Algae are the producers, small fish are the first consumers and herons are the second consumers.
2. Algae to small fish: 10 percent of 5000 = 500 units.
3. Small fish to herons: 10 percent of 500 = 50 units.
4. The rest is used by each organism for living (moving, growing, keeping warm) and is given off as heat, so it is not passed on.

About 50 units reach the herons; the rest is used for life processes or lost as heat.

Questions

1. Write the process of photosynthesis as a word equation.

2. Cells release the energy stored in food by cellular respiration. Which two substances do most cells need to take in for this, and which two are made?

3. Producers in a pond capture 8000 units of energy. About 10 percent is passed on at each step. How much energy reaches the first consumers?

4. Why are food chains usually only three to five steps long?

5. Corals in warm seas have tiny algae living inside them. The algae make food for the coral, and the coral gives the algae shelter. What kind of relationship is this?

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6. Two kinds of birds in a mangrove forest both eat the same small fish. Which type of interaction is this, and what may happen if the fish become fewer?

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7. An oryx reserve has a fixed area, a limited amount of grass and a limited amount of water. Why does the oryx population stop growing after a time?

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8. Why is an ecosystem with many different species usually more stable than one with only a few?

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9. The carbon atoms in a fallen mangrove leaf can later be part of a crab. Describe how matter is cycled in an ecosystem to make this possible.

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10. Fossils of sea shells are found high in the Hajar mountains. What does this evidence tell scientists about the place long ago?

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11. The arm bones of a human, the wing bones of a bat and the flipper bones of a whale have a similar pattern, though the limbs do different jobs. What do scientists conclude from this evidence?

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12. Some lizards in a sandy desert are light brown and some are dark. Birds find the dark lizards easier to see. Explain, using natural selection, how the group of lizards may change over many generations.

Section 2: Earth and space

Standards practised: MS-ESS1-1, MS-ESS1-2, MS-ESS1-3, MS-ESS1-4

A worked example

Three layers of sedimentary rock lie flat, one on top of another and have not been disturbed. Layer X is at the top and has leaf fossils. Layer Y is in the middle and has shell fossils. Layer Z is at the bottom and has fish fossils. Which layer is the oldest, and which fossils are the youngest?

1. In undisturbed layers, new sediment is laid down on top of the older sediment.
2. So the bottom layer, Z, is the oldest and the top layer, X, is the youngest.
3. Fossils are the same age as the layer they are found in, so the fish fossils are the oldest and the leaf fossils are the youngest.

Layer Z is the oldest; the leaf fossils in layer X are the youngest.

Questions

1. What causes the seasons on Earth?

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2. Summer days in the UAE are long and hot. Explain why, using the Sun's position and the tilt of the Earth.

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3. Some people think it is hotter in summer because Earth is closer to the Sun. Is this correct? Give a reason.

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4. The Moon takes about 29.5 days to go through all its phases. Why does its shape seem to change?

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5. During a solar eclipse, which object is in the middle, and during a lunar eclipse, which object is in the middle?

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6. There is not a solar eclipse every month, even though the Moon passes between the Earth and the Sun each month. Why not?

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7. What force keeps the planets moving in orbits around the Sun, instead of travelling off in straight lines?

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8. The Hope Probe took about seven months to travel from Earth and enter orbit around Mars in February 2021. Which planet is Mars counting outwards from the Sun, and is it a rocky planet or a gas giant?

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9. Light travels at about 300 000 km per second. The Sun is about 150 000 000 km from Earth. How many seconds does sunlight take to reach Earth, and about how many minutes is that?

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10. Put these in order from smallest to largest: the Milky Way galaxy, the Sun, Earth, the solar system.

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11. A cliff shows horizontal layers of rock. A crack (a fault) cuts straight through all of the layers. Is the fault older or younger than the layers? Explain.

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12. Earth is about 4.6 billion years old, but people have lived on it for only a very small part of that time. Explain how rock layers and fossils help scientists build a timeline of Earth's history.

Section 3: Matter, forces and energy

Standards practised: MS-PS4-1, MS-PS4-2, MS-PS4-3, MS-PS2-2, MS-PS2-3

A worked example

A family car travels 210 km in 2.5 hours. What is its average speed? If it keeps the same average speed, how far does it travel in 4 hours?

1. Speed = distance ÷ time.
2. 210 km ÷ 2.5 hours = 84 km per hour.
3. Distance in 4 hours = speed × time = 84 × 4 = 336 km.

84 km per hour; 336 km in 4 hours.

Questions

1. A wave has a high crest and a deep trough. Which property of the wave does this describe, its amplitude or its wavelength? What does a larger amplitude mean for a sound wave and for a water wave?

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2. A sound wave has a frequency of 500 hertz and a wavelength of 0.68 m. Use speed = frequency × wavelength to find the speed of the sound.

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3. Why can astronauts not hear each other speak outside their spacecraft, even if they stand close together?

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4. A ray of light hits a flat mirror at an angle of 30 degrees to the normal (the line at right angles to the mirror). At what angle does it reflect?

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5. List these waves in order from the longest wavelength to the shortest: X-rays, visible light, radio waves, ultraviolet.

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6. Why do people in the UAE wear sunscreen and sunglasses when they spend a long time in the midday sun?

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7. A digital signal is made of a code of 0s and 1s. Give one reason why digital signals are used to send pictures and measurements from the Hope Probe to Earth.

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8. Name the energy change that happens in a solar panel, and say why solar parks suit the UAE.

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9. Two magnets are placed with their north poles facing each other. What do you feel when you try to push them together? Name one way to make an electromagnet stronger.

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10. A cyclist rides 36 km in 2 hours. A runner covers 10 km in 1 hour. Who has the greater average speed, and by how much?

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11. A car starts from rest and reaches 20 m per second after 5 seconds. What is its acceleration? Use acceleration = change in speed ÷ time.

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12. A force of 3000 N pushes a car of mass 1000 kg. Use force = mass × acceleration to find the acceleration. Then explain why a passenger should wear a seatbelt when the car stops suddenly.

Answer guide for parents

Section 1: Living things

1. Carbon dioxide + water, using energy from light, makes glucose (sugar) + oxygen.
2. They take in glucose (from food) and oxygen. Carbon dioxide and water are made, and energy is released.
3. 800 units (10 percent of 8000)
4. Only about 10 percent of the energy is passed on at each step, so after a few steps there is too little energy left to support another group of animals.
5. Mutualism, because both organisms benefit.
6. Competition. If the fish become fewer, each kind of bird gets less food, so their numbers may fall.
7. Limiting factors such as food, water and space set a limit (the carrying capacity) on how many oryx the area can support.
8. With many species there are many food sources and many links in the food web, so if one species decreases, others can still survive and the ecosystem keeps working.
9. Decomposers break down the dead leaf and return carbon to the soil and air. Plants take carbon in again (as carbon dioxide) and build it into new leaves, which animals eat. The atoms are reused, not lost.
10. That the land was once covered by sea. The shells were buried in sediment under the water, and later the rock was lifted up.
11. The similar bone structure is evidence that these animals share a common ancestor and have changed over very long periods of time.
12. Light brown lizards are less often eaten, so more of them survive and have young. They pass the light colour on, so over many generations a larger share of the group is light brown.

Section 2: Earth and space

1. Earth's axis is tilted (about 23.5 degrees) as it travels around the Sun. The tilt changes how directly sunlight reaches each part of Earth during the year.
2. In summer our part of Earth is tilted towards the Sun. The Sun is higher in the sky, so its light is more direct and the days have more hours of daylight.
3. No. The tilt of Earth's axis causes the seasons. (Earth is actually slightly farther from the Sun in the northern summer.)
4. The Moon always has one half lit by the Sun. As it orbits Earth we see different amounts of the lit half.
5. Solar eclipse: the Moon is between the Sun and Earth. Lunar eclipse: Earth is between the Sun and the Moon.
6. The Moon's orbit is tilted compared with Earth's orbit, so most months the Moon passes a little above or below the line between the Sun and Earth.
7. The Sun's gravity pulls on the planets. Gravity is stronger for objects with more mass and weaker when the distance is greater.
8. Mars is the fourth planet from the Sun. It is a rocky planet.

9. 150 000 000 ÷ 300 000 = 500 seconds, which is about 8 minutes (8 minutes 20 seconds).
10. Earth, the Sun, the solar system, the Milky Way galaxy.
11. Younger. The layers had to be there first before the crack could cut through them.
12. Layers show the order in which events happened (oldest at the bottom). Fossils show which living things existed at each time and which environments they lived in. Together they give a record of changes over billions of years.

Section 3: Matter, forces and energy

1. Its amplitude (the height of the wave from its rest position). A larger amplitude means more energy: a louder sound, or a taller and more powerful water wave.
2. 500 × 0.68 = 340 m per second
3. Sound is a mechanical wave and needs matter (a medium such as air) to travel through. Space is almost empty, so sound cannot travel.
4. 30 degrees (the angle of reflection equals the angle of incidence).
5. Radio waves, visible light, ultraviolet, X-rays.
6. The Sun gives out ultraviolet waves, which carry enough energy to harm skin and eyes. Sunscreen and sunglasses block or absorb much of it.
7. A digital signal can be copied and sent without losing detail, and errors can be found and corrected, so the information arrives clearly even over a very long distance. (The signals travel as radio waves.)
8. Light energy is changed into electrical energy. The UAE has many hours of strong sunshine and a lot of open land.
9. They push apart (like poles repel). An electromagnet gets stronger with more turns of wire, a larger electric current or an iron core.
10. The cyclist: 36 ÷ 2 = 18 km per hour. The runner: 10 km per hour. The cyclist is faster by 8 km per hour.
11. 20 ÷ 5 = 4 m per second squared (its speed goes up by 4 m per second every second).
12. Acceleration = 3000 ÷ 1000 = 3 m per second squared. When the car stops suddenly the passenger's body keeps moving forward at the same speed (inertia), so a seatbelt provides the force that stops the body safely.